PROJECT SYNOPSIS & ARCHITECTURE

Adaptive Spatial Intelligence and Predictive Dispatch Optimization for Last-Mile Delivery Operations

The increasing demand for fast and reliable last-mile delivery requires logistics systems to efficiently manage dynamic customer demand, inventory, and delivery resources. Conventional approaches based on static service regions and reactive planning often lead to inefficient operations and resource imbalance.

This project proposes an integrated framework that combines adaptive spatial partitioning, demand forecasting, dispatch optimization, and driver assignment to support intelligent delivery planning. An interactive simulation engine further demonstrates the end-to-end operational workflow, providing a platform for evaluating the proposed methodologies under realistic delivery scenarios.

Layer 1 – Adaptive Spatial Partitioning

Partition the operational service region into adaptive delivery zones that accurately represent the spatial distribution of customer demand while maintaining balanced workloads. The objective is to merge low-demand regions and split high-demand hotspots to create operational zones suitable for forecasting, dispatch, and driver assignment.

Customer orders were first aggregated at the H3 Resolution-7 hexagonal grid to obtain the spatial demand of each cell:

$$D_i = \sum_{k=1}^{N_i} 1$$

where D_i denotes the demand of H3 cell i, and N_i is the number of customer orders within the cell.

The demand distribution was then analyzed to compute adaptive sparse and hotspot thresholds,

$$T_s = P_{25}(D), \qquad T_h = P_{99}(D)$$

which served as the initialization criteria for the greedy merge–split zoning process. Cells with demand below T_s were marked for merging, while cells exceeding T_h were identified as hotspot candidates for recursive subdivision. Cells with demand below T_s were greedily merged with the neighboring cell having the highest demand, thereby reducing sparse operational regions while preserving local demand density. Conversely, cells exceeding T_h were recursively subdivided into finer H3 resolutions until the demand distribution became balanced. This greedy merge–split strategy iteratively minimized spatial imbalance while maintaining the hierarchical structure of the H3 grid, resulting in adaptive operational zones suitable for downstream forecasting and optimization.

The final zone mapping is represented as

$$Z = f(H, D)$$

where H denotes the hierarchical H3 grid and D represents the spatial demand distribution.

ALGORITHMS USED:

H3 Spatial Indexing ▪ Percentile-Based Thresholding ▪ Greedy Neighbor Merging ▪ Recursive Hotspot Splitting ▪ Adaptive Zone Construction

Layer 2 – Demand Forecasting

Predict the customer demand for the upcoming operational shift at the H3-cell level and aggregate these predictions to adaptive operational zones, enabling proactive inventory planning and dispatch optimization.

Historical order demand was aggregated at the H3 Resolution-8 level for each operational shift. A Long Short-Term Memory (LSTM) network was trained to learn temporal demand patterns using sequential historical observations.

For an LSTM cell,

$$\begin{aligned}
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\
\tilde{C}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
h_t &= o_t \odot \tanh(C_t)
\end{aligned}$$

where x_t denotes the input feature vector, h_t the hidden state, and C_t the memory cell. The model outputs the predicted demand $\hat{D}_i$ for each H3 cell.

Predicted H3-cell demands were aggregated into adaptive operational zones using weighted averaging based on historical cell importance:

$$\hat{D}_Z = \frac{\sum_{i=1}^{n} w_i \hat{D}_i}{\sum_{i=1}^{n} w_i}$$

where $\hat{D}_i$ is the predicted demand of H3 cell i, w_i is its rolling historical weight, and $\hat{D}_Z$ is the forecasted demand for operational zone Z. The resulting zone-level forecasts are supplied to the dispatch optimization layer for proactive decision-making.

ALGORITHMS USED:

Long Short-Term Memory (LSTM) ▪ Time-Series Forecasting ▪ Weighted Zone Aggregation

Sublayer 3.1 – Priority-Based Batching

Group pending products into feasible dispatch batches by prioritizing urgent products while preventing indefinite carry-over, subject to dispatch-center inventory, destination zone, and vehicle-capacity constraints.
Each product was assigned an Expected Priority Score (EPS) that combines its operational priority with the penalty incurred from repeated retention:

$$\begin{aligned}\text{EPS} &= \frac{\text{Priority Score}}{\text{Carry-over Penalty}} \\ &= \alpha(\text{Product Priority Score}) + \beta(\text{Carry-over Penalty})\end{aligned}$$

where the Product Priority Score reflects the operational importance of the product, while the Carry-over Penalty increases after every retained dispatch cycle, ensuring long-pending products gradually receive higher priority. Products were ranked in descending order of EPS and greedily grouped into candidate batches while satisfying inventory availability, destination-zone compatibility, and vehicle-capacity constraints. The generated batches formed the feasible solution space for the dispatch optimization stage.

ALGORITHMS USED:

Expected Priority Score (EPS) ▪ Priority Ranking ▪ Constraint-Based Batch Formation

Sublayer 3.2 – BUS-Based Dispatch Optimization

Problem: Select the optimal combination of candidate batches that maximizes overall operational efficiency by balancing immediate demand, future demand, storage utilization, dispatch urgency, and fleet utilization.
Each candidate batch was evaluated using the Batch Utility Score (BUS):

$$\text{BUS} = w_1\text{BP} + w_2\text{VU} + w_3\text{SU} + w_4\text{RC} + w_5\text{FO}$$

where:

- **BP** – Batch Priority (aggregate product priority)
- **VU** – Vehicle Utilization
- **SU** – Storage Urgency at the dispatch center
- **RC** – Retention Cost incurred by delaying dispatch
- **FO** – Forecast Opportunity representing the benefit of retaining inventory for predicted future demand

The optimal dispatch schedule was obtained by solving a Binary Integer Optimization problem:

$$\max \sum_{i=1}^{m} \text{BUS}_i x_i$$

subject to inventory, vehicle-capacity, and fleet-availability constraints, where $x_i \in \{0, 1\}$ indicates whether candidate batch i is selected for dispatch.

ALGORITHMS USED:

Batch Utility Score (BUS) ▪ Binary Integer Optimization ▪ Fleet-Constrained Dispatch Scheduling

Layer 4 – Driver Assignment Optimization

Assign available delivery drivers to operational zones such that the overall driver familiarity with their assigned zones is maximized while ensuring a one-to-one assignment between drivers and zones.

A familiarity score was computed for every driver–zone pair using the driver's historical service frequency within the H3 cells belonging to each operational zone.

$$F_{ij} = \sum_{k \in Z_j} f_{ik}$$

where:

- F_{ij} = familiarity score of driver i for zone j
- f_{ik} = historical service frequency of driver i in H3 cell k
- Z_j = set of H3 cells belonging to operational zone j

The resulting familiarity matrix was transformed into a cost matrix for optimization:

$$C_{ij} = F_{\max} - F_{ij}$$

where $F_{\max}$ is the maximum familiarity score in the matrix.

The Hungarian Algorithm was applied to the cost matrix to determine the globally optimal one-to-one mapping between available drivers and operational zones by minimizing the total assignment cost, which is equivalent to maximizing the total familiarity score.

$$\min \sum_{i=1}^{n} \sum_{j=1}^{m} C_{ij} x_{ij}$$

subject to

$$\sum_{j} x_{ij} = 1, \qquad \sum_{i} x_{ij} = 1, \qquad x_{ij} \in \{0, 1\}$$

where $x_{ij} = 1$ if driver i is assigned to zone j, and 0 otherwise.

ALGORITHMS USED:

Driver Familiarity Scoring ▪ Cost Matrix Construction ▪ Hungarian Assignment Algorithm

Layer 5 – Interactive Simulation Environment

Provide an event-driven simulation environment that integrates all backend optimization layers into a unified platform for evaluating delivery operations under realistic scenarios. The simulation enables users to visualize the complete order fulfillment workflow from order placement to final delivery.

Interactive Order Generation: Allows users to place customer orders by selecting delivery locations, products, and quantities through an interactive interface.

Real-Time Zone Construction: Dynamically generates adaptive operational zones using the AHSI layer based on the current spatial demand distribution.

Demand Forecast Integration: Predicts upcoming zone-level demand using the trained forecasting model and supplies these forecasts to the dispatch optimization layer.

Dispatch Decision Visualization: Simulates product batch formation, dispatch scheduling, and inventory movement across dispatch centers.

Driver Assignment & Delivery Tracking: Assigns available drivers to operational zones and visualizes delivery execution, driver movement, and order completion in real time.

Operational Dashboard: Displays key performance metrics including active orders, dispatch-center inventory, driver availability, zone-wise demand, dispatch status, and delivery progress throughout the simulation.

The simulation engine serves as an end-to-end validation platform, demonstrating the interaction between adaptive zoning, demand forecasting, dispatch optimization, and driver assignment within a unified operational workflow.